CAPSTONE EVALUATION FORM

HITOVIS: A Smart IoT-Based Catfish Aquaculture Management System

Name: _____

☐ Community/Fisherman ☐ Client ☐ IT Expert ☐ Others

Direction: Respond to the following statement;the number of your responses corresponds using the scale below.

Numerical Value	Verbal Description
5	Strongly Agree
4	Agree
3	Moderately Agree
2	Disagree
1	Strongly Disagree

IoT DEVICE AND SYSTEM QUALITY EVALUATION (ISO/IEC 25010:2011)

	5	4	3	2	1
1. FUNCTIONAL SUITABILITY					
1.1 Functional completeness. The device provides the functions needed for the farm activities it is intended to support.					
1.2 Functional correctness. The information produced by device is correct for the tasks evaluated.					
1.3 Functional appropriateness. The device helps users complete intended farm activities without unnecessary steps.					
1.4 As a whole, the device provides appropriate and correct functions that meet the needs of its intended farm activities.					
2. RELIABILITY					
2.1 Maturity. The device operates consistently during normal use.					
2.2 Availability. The device is available when needed for intended farm operations.					
2.3 Fault tolerance. When a component or connection has a problem, device continues the functions designed to remain available.					
2.4 Recoverability. After an interruption, the device restores access to previously saved information needed for operation.					
2.5 As a whole, the device performs reliably under intended operating conditions and supports recovery after interruptions.					
3. PERFORMANCE EFFICIENCY					
3.1 Time behaviour. The device responds within a reasonable time during normal use.					
3.2 Resource utilization. The device uses available computing and network resources efficiently.					

3.3	Capacity. The device handles the amount of information and number of users required for its intended use.					
3.4	As a whole, the device performs its intended tasks within a reasonable time while using available resources efficiently.					
4. USABILITY						
4.1	Appropriateness recognizability. The purpose of the device and its functions is clear to users.					
4.2	Learnability. Users can learn to use the device with the available instructions and assistance.					
4.3	Operability. The device controls are easy to operate for intended farm tasks.					
4.4	User error protection. The device helps users avoid or correct mistakes before they cause unwanted results.					
4.5	User interface aesthetics. The visual presentation of the device is clear and orderly.					
4.6	Accessibility. The device can be used by intended users with different needs and abilities.					
4.7	As a whole, the device is easy for intended users to understand, learn, and operate.					
5. SECURITY						
5.1	Confidentiality. Farm information is accessible only to users permitted to view it.					
5.2	Integrity. Farm records are protected from changes by users not permitted to edit them.					
5.3	Non-repudiation. Important device actions can be verified from retained evidence if their occurrence is questioned.					
5.4	Accountability. Recorded actions can be linked to the responsible user or device.					
5.5	Authenticity. The device checks a user's identity before allowing access to protected functions.					
5.6	As a whole, the device protects farm information and restricts access and actions to authorized users.					
6. COMPATIBILITY						
6.1	Co-existence. The device works alongside other applications in the intended environment without disrupting their normal operation.					
6.2	Interoperability. The device exchanges usable information correctly between its connected application and device components.					
6.3	As a whole. The device works appropriately with the other components and applications needed for its operation.					
7. MAINTAINABILITY						
7.1	Modularity. A change to one device component can be made without unnecessarily affecting other components.					
7.2	Reusability. Existing device components can support related functions without being recreated.					

7.3	Analyzability. Available technical information helps identify the cause of a device problem.					
7.4	Modifiability. The device can be corrected or updated without causing unrelated functions to stop working.					
7.5	Testability. The device provides enough information to check whether its functions produce expected results.					
7.6	As a whole, the device can be analyzed, tested, maintained, and updated without unnecessarily disrupting other functions.					
8. PORTABILITY						
8.1	Adaptability. The device can be adapted to the supported devices and operating environments needed by the farm.					
8.2	Installability. The device can be set up successfully in its intended operating environment.					
8.3	Replaceability. A supported device version can replace a previous version while retaining the functions needed by the farm.					
8.4	As a whole, the device can be adapted, installed, and replaced within its supported operating environments.					

Unified Theory of Acceptance and Use of Technology (UTAUT)

1. PERFORMANCE EXPECTANCY	5	4	3	2	1
1.1 Using the system would help me monitor tank conditions more effectively.					
1.2 Using the system would help me complete my assigned aquaculture tasks more quickly.					
1.3 Using the system would help me make informed decisions about tank management.					
1.4 Using the system would improve my effectiveness in managing catfish production.					
1.5 As a whole, the system would help me perform my catfish aquaculture tasks more effectively.					
2. EFFORT EXPECTANCY					
2.1 Learning to use the system would be easy for me.					
2.2 My interaction with the system would be clear and understandable.					
2.3 It would be easy for me to become skillful at using the system.					
2.4 I would find the system easy to use for tasks assigned to my role.					
2.5 As a whole, the system would be easy for me to learn and use.					
3. SOCIAL INFLUENCE					
3.1 People who influence how I perform farm tasks would think that I should use the system.					
3.2 People whose opinions I value would encourage me to use the system.					

3.3	The farm owner or supervisor would support my use of the system.					
3.4	As a whole, people whose opinions I value would support my use of the system.					
4. FACILITATING CONDITIONS						
4.1	I have access to the resources needed to use the system, such as a suitable device and internet connection.					
4.2	I have the knowledge needed to use the system functions assigned to my role.					
4.3	The system is compatible with the other tools I use for my farm tasks.					
4.4	I can obtain assistance from a designated person when I have difficulty using the system.					
4.5	As a whole, I have the support needed to use the system.					
5. BEHAVIORAL INTENTION						
5.1	I intend to use the system for my assigned farm tasks when it is available.					
5.2	I plan to use the system regularly during the next production cycle if I am given access.					
5.3	I expect to continue using the system for relevant farm tasks in the future.					
5.4	As a whole, I intend to use the system regularly for my assigned aquaculture tasks.					

Comments/Suggestion:_____

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SIGNATURE